

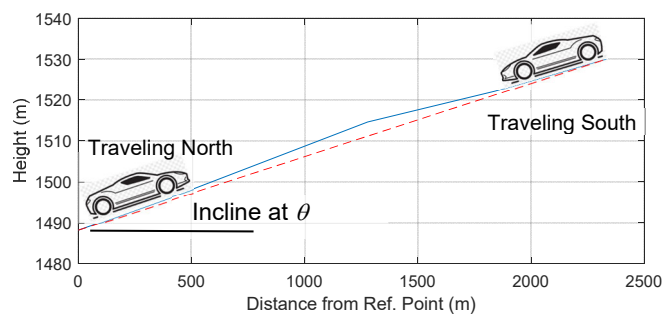
MeEn 335 System Dynamics

Lab: Modeling Car Coast Down work in assigned groups of two



How does the amount of power required to drive a car change at different speeds? The goal of this lab is to create a simulation of a car coasting freely on a slight incline, and to compare the simulation with measurements in order to determine the drag coefficient of the car and the coefficient of kinetic friction during rolling. These will be used to estimate the amount of power required to drive the vehicle at 55mph and 100 mph.

Measurements were taken from a 2016 Lexus GS350 as it rolled either down (traveling South) or up (traveling North) a relatively steady incline that can be approximated as having a constant angle $\theta = 1.03$ degrees. During all of the measurements, a steady wind was blowing, which the weather service reported as coming from the North at approximately 7mph.



The python file “CarCoastData_plotdata.py” loads the data that was measured and plots it and contains notes about the variable names. The variables of interest are “t1” time in seconds, “v1” velocity in m/s, both for the vehicle traveling South and “t2” and “v2” for the vehicle traveling North. Note that all of these measurements are subject to the measurement uncertainties inherent in using a cell phone to determine and record GPS coordinates.

You will build two models of the car’s motion, a linear model, and a nonlinear model.

Linear Model

Our first step will be to model the system as if it were linear, meaning that the velocity of the car is specified by this first-order linear differential equation:

$$m\dot{v} + bv = f(t)$$

The mass of the car is 1760 kg plus 150 kg of passengers. The applied force $f(t)$ is the constant force due to gravity. Using both the uphill and downhill motion (see Speed vs. Time plot), fit the parameter b to see how close you can get your model to match the data. Generate a plot comparing the predicted and observed

behaviors.

Nonlinear Model

While the linear model is a good approximation of the vehicle over a small range of speeds, we can improve this model using a nonlinear model that accounts for aerodynamic drag and speed-independent friction losses. Find the differential equation governing the motion of the car including the effect of gravity, aerodynamic drag and friction losses.

Use the aerodynamic law $F_a = \frac{1}{2}C_d A_p \rho_f v^2$, where C_d is the (unknown) drag coefficient, A_p is the projected area of the vehicle (which you can assume to be $A_p = 2.15 \text{ m}^2$), $\rho_f = 1.06 \text{ kg/m}^3$ is the density of the fluid and v is the speed of the vehicle relative to the wind. So, if the vehicle is moving North into the wind $v = v_{car} + v_{wind}$ and if the vehicle is moving south $v = v_{car} - v_{wind}$.

Model the friction losses using the Coulomb friction law, $F_f = \mu_k N$ where N is the force normal to the road and μ_k is the unknown coefficient of kinetic friction for the rolling vehicle.

Once you have derived the equation of motion, use `solve_ivp` in python (or `ode45` in matlab) to simulate the response of the vehicle and compare the velocity as a function of time with the GPS measurements. Then, iterate on the unknown parameters C_d and μ_k until the simulations agree with measurements.

- What effect does each term, i.e those due to C_d and μ_k , have on the velocity versus time? To what extent can you tell their effects apart?

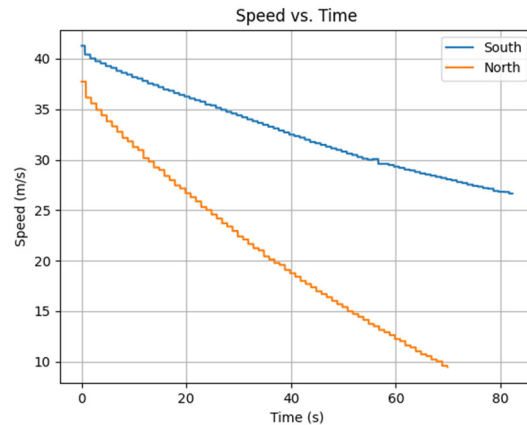
Using Your Model to Predict Future Behavior

Use the values of C_d and μ_k that you have found to compute the power required for the vehicle to travel 55 mph and 100 mph on level ground. You will probably find it easiest to compute the power in metric units (i.e. kW) but report your answer in horsepower so one can easily see what fraction of the engine's maximum horsepower is being used.

Uncertainty Analysis

Once you have an estimate of the parameters C_d and μ_k , perform additional analysis to quantify the uncertainty in these parameters. Your goal is to understand how accurate each of these parameters is. Some questions that you may explore include: Could I obtain about the same level of agreement with different values for these parameters, or different combinations of them? What is the largest change in each parameter that would still produce an acceptable fit to the data? If we repeated the test, how much might these values vary? What effect will all of this have on your horsepower estimates? Your analysis and results in this section should demonstrate that you understand the dynamics of the vehicle and how these parameters affect its response.

Design report: Hand in a professionally formatted report in **PDF format** summarizing your analysis and results. Please limit your report to no more than **three (3)** pages in length. You should include detailed derivations (handwritten ok) and Python/Matlab codes in an appendix, which can be as long as you would like, but your reader should be able to understand what you have done and the results of your analysis



without having to turn to the appendix. (i.e. your final equation of motion should be given within the body of the report, as well as at least one figure showing your key results.) The key items included in your report should include:

- Final nonlinear equation of motion for the car. Don't include the full derivation, just the result and explain to the reader what each term means and how you know that its sign/units/etc are correct.
- Plot showing the behavior of the linear model compared to the data
- One or more plots showing the behavior of the nonlinear model compared to the data
- A table such as this summarizing your results:

Damping coefficient b for the linear model	
Drag Coefficient, C_d , and uncertainty estimate	
Coefficient of kinetic friction during rolling, μ_k , and uncertainty estimate	
Power required at 55mph (reported in horsepower)	
Power required at 00mph (reported in horsepower)	

- Justification for the uncertainty estimates given above, and the procedure that you used to quantify them. This should include calculations, figures, discussion, etc... and not simply a list of things that you don't know.

Rubrik:

- Accuracy of results: 40%
- Professionalism of report and effective use of figures: 20%
- Clear and accurate explanations of the dynamics involved and the uncertainty estimates: 40%.